Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C034 **BMH1** 5 genes / 3 species 6 direct pairs	**Y1 BMH1** GenAge 1033 **GenAge annotation** **Unclear** Lifespan: Increase + Decrease Refs: 19805817, 20657825	**W1 ftt-2** GenAge 288 **GenAge annotation** **Pro-Longevity** Lifespan: Increase + Decrease Refs: 16860373, 18423931	**F1 14-3-3epsilon** GenAge 831 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18665908	No gene in this candidate network
	Y2 BMH2 GenAge 1670 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 19805817	**W2 par-5** GenAge 493 **GenAge annotation** **Pro-Longevity** Lifespan: Increase + Decrease Refs: 16860373, 18423931		
Direct pairs: W1–Y1 (n:m); W1–Y2 (n:m); W2–Y1 (n:m); W2–Y2 (n:m); F1–Y1 (1:m); F1–Y2 (1:m)				
C017 **CAT5** 3 genes / 3 species 3 direct pairs	**Y1 CAT5** GenAge 1436 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 17492370, 18340043	**W1 clk-1** GenAge 130 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 8638122, 10202142 12709403, 15269213	No gene in this candidate network	**M1 Coq7** GenAge 136 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 16195414, 25106098
Direct pairs: W1–M1 (1:1); W1–Y1 (1:1); M1–Y1 (1:1)				
C012 **CTA1** 3 genes / 3 species 3 direct pairs	**Y1 CTA1** GenAge 1694 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 20696905	No gene in this candidate network	**F1 Cat** GenAge 91 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 8108730	**M1 Cat** GenAge 92 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 15879174, 16144468
Direct pairs: F1–M1 (1:1); F1–Y1 (n:1); M1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.